

99.500 Applied Mathematics for Engineering Spring Semester AY2026

Bessel Ordinary Differential Equations

Week 5 Outline

- Bessel ODEs and Bessel functions
- Diffusive reaction in a cylindrical shaped catalyst

Bessel ODEs

$$\text{Bessel ODE: } x^2 y'' + xy' + (x^2 - p^2)y = 0$$

Solution:

$$y = c_1 J_p(x) + c_2 Y_p(x)$$

$J_p(x)$: Bessel function of the first kind of order p and argument x

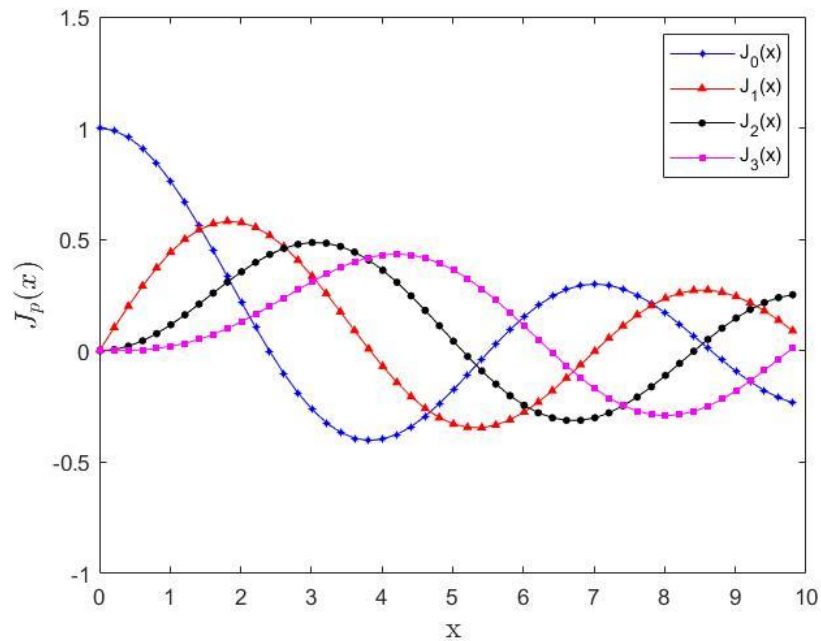
$$J_p(x) = \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{1}{2}x\right)^{2n+p}}{n! \Gamma(n+p+1)}$$

$Y_p(x)$: Bessel function of the second kind of order p and argument x

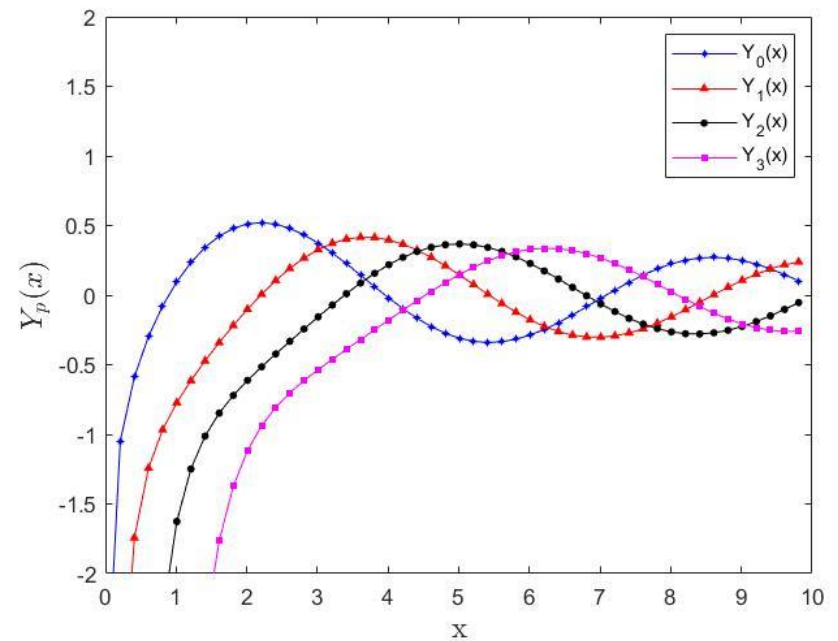
$$Y_p(x) = \frac{\cos(\pi p) J_p(x) - J_{-p}(x)}{\sin(\pi p)}$$

Bessel Functions

Matlab: `besselj(p,x)`



Matlab: `bessely(p,x)`



Modified Bessel ODEs

$$\text{Modified Bessel ODE: } x^2 y'' + xy' - (x^2 + p^2)y = 0$$

Solution:

$$y = c_1 I_p(x) + c_2 K_p(x)$$

$I_p(x)$: Modified Bessel function of the first kind of order p and argument x

$$I_p(x) = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}x\right)^{2n+p}}{n!\Gamma(n+p+1)} = i^{-p} J_p(ix)$$

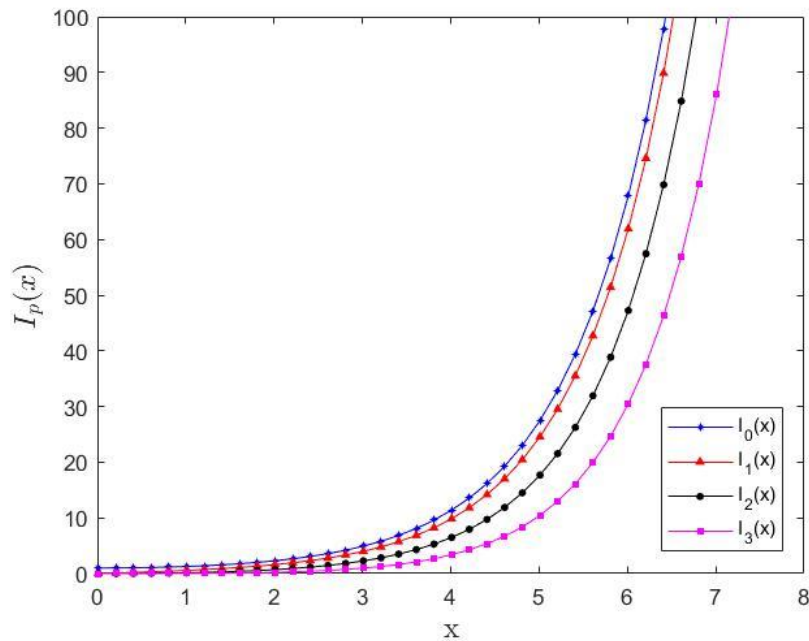
$K_p(x)$: Modified Bessel function of the second kind of order p and argument x

$$K_p(x) = \frac{\pi}{2} \left[\frac{I_{-p}(x) - I_p(x)}{\sin(\pi p)} \right]$$

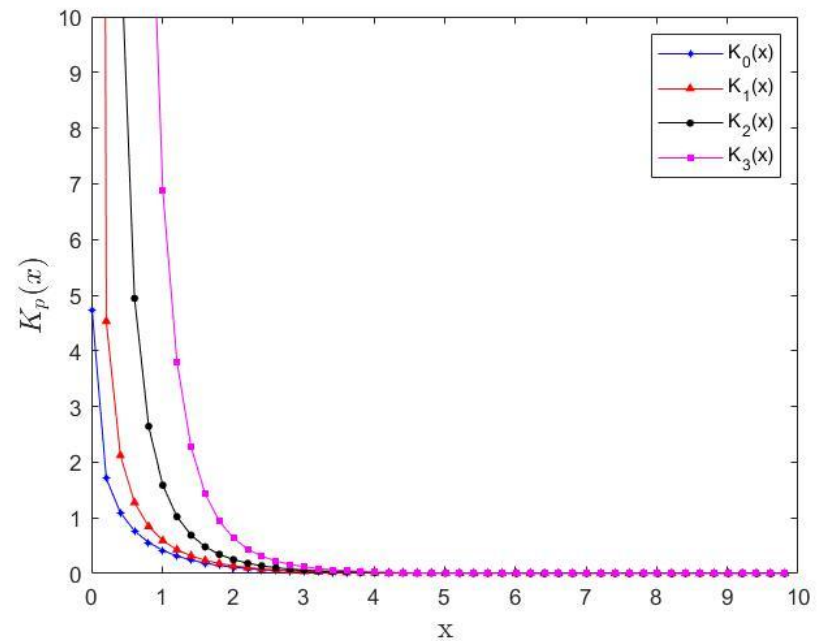
Where does Bessel ODEs arise?

Bessel Functions

Matlab: `besseli(p,x)`



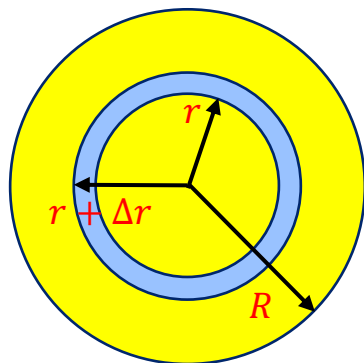
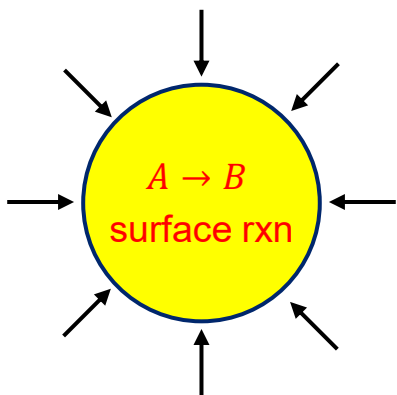
Matlab: `besselk(p,x)`



Problem generating Bessel ODEs

Bessel ODEs naturally arise in problem with radial symmetry

Example: Diffusive reaction in a cylindrical shaped catalyst



In: $\left(-D \frac{dC_A}{dr} 2\pi r L \right) \Big|_r$

Out: $\left(-D \frac{dC_A}{dr} 2\pi r L \right) \Big|_{r+\Delta r}$

Generation: 0

Consumption: $-r_A a \Delta V = -k C_A a (2\pi r \Delta r L)$

Specific surface area: $a \left(\frac{\text{m}^2}{\text{m}^3} \right)$ for catalyst

Accumulation: 0 (assuming steady-state)

$$\left(-D \frac{dC_A}{dr} 2\pi r L \right) \Big|_r - \left(-D \frac{dC_A}{dr} 2\pi r L \right) \Big|_{r+\Delta r} - k C_A a (2\pi r \Delta r L) = 0$$

$$\text{BC: } \begin{cases} \frac{dC_A}{dr} \Big|_{r=0} = 0 \\ \left(-D \frac{dC_A}{dr} 2\pi r L \right) \Big|_{r=R} = [k_c (C_A - C_{A,0}) 2\pi r L]_{r=R} \end{cases}$$

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dC_A}{dr} \right) - \frac{k a}{D} C_A = 0 \xrightarrow{?} x^2 y'' + x y' - (x^2 + p^2) y = 0$$

Nondimensionalization

$$y = \frac{C_A}{C_{A,0}} \quad x = \sqrt{\frac{ka}{D}} r = \frac{r}{L_d} \quad \text{where dynamic length } L_d = \sqrt{\frac{D}{ka}}$$

$$\begin{aligned} \frac{1}{r} \frac{d}{dr} \left(r \frac{dC_A}{dr} \right) - \frac{ka}{D} C_A &= 0 \quad \Rightarrow \quad \frac{1}{L_d x} \frac{d}{d(L_d x)} \left(L_d x \frac{d(C_{A,0} y)}{d(L_d x)} \right) - L_d^{-2} C_{A,0} y = 0 \\ &\Rightarrow \quad x^2 y'' + x y' - x^2 y = 0 \quad \text{with } p = 0 \end{aligned}$$

$$\left. \frac{dC_A}{dr} \right|_{r=0} = 0 \quad \Rightarrow \quad y'(0) = 0$$

$$\begin{aligned} \left(-D \frac{dC_A}{dr} 2\pi r L \right) \Big|_{r=R} &= [k_c (C_A - C_{A,0}) 2\pi r L]_{r=R} \\ \Rightarrow -y'(\alpha) &= \beta [(y(\alpha) - 1)] \end{aligned}$$

$$\alpha = \frac{R}{L_d} = R \sqrt{\frac{ka}{D}} \quad \& \quad \beta = \frac{k_c}{\sqrt{kaD}}$$

Solution

$$x^2 y'' + xy' - x^2 y = 0$$

Modified Bessel ODE with $p = 0$

$$y(x) = c_1 I_0(x) + c_2 K_0(x)$$

$$y'(0) = 0 \implies c_1 I_1(0) - c_2 K_1(0) = 0$$

$$\implies c_2 = 0 \text{ \& } y(x) = c_1 I_0(x)$$

$$-y'(\alpha) = \beta [(y(\alpha) - 1)] \implies -c_1 I_1(\alpha) = \beta [(c_1 I_0(\alpha) - 1)]$$

$$\implies c_1 = \frac{\beta}{\beta I_0(\alpha) + I_1(\alpha)}$$

$$\implies y(x) = \frac{\beta I_0(x)}{\beta I_0(\alpha) + I_1(\alpha)}$$

What happens if the derived model:

$$x^2 y'' + xy' - x^2 y = qx^2$$

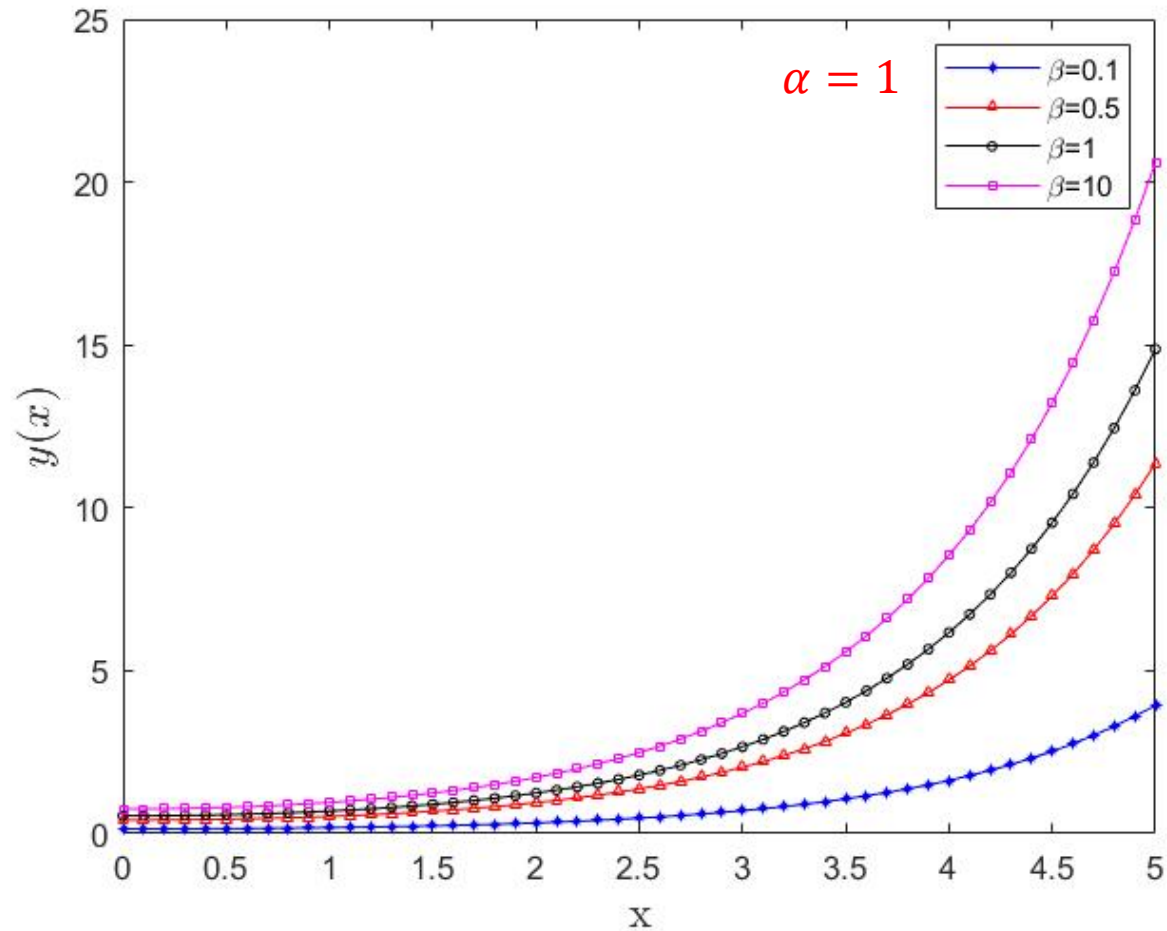
Let $z = y + q$

$$x^2 z'' + xz' - x^2 z = 0$$

$$z(x) = c_1 I_0(x) + c_2 K_0(x)$$

$$y(x) = c_1 I_0(x) + c_2 K_0(x) - q$$

Dimensionless Concentration Profile



Catalyst Efficiency

Define v : actual reaction rate in the catalyst

$$v = \left(D \frac{dC_A}{dr} 2\pi r L \right)_{r=R} = \frac{2\pi R L D C_{A,0}}{L_d} \left(\frac{dy}{dx} \right)_{x=\alpha}$$

Define γ : reaction rate without diffusion resistance in the catalyst

$$\gamma = k C_{A,0} a \pi R^2 L$$

$$y'(\alpha) = \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)} \implies v = \frac{2\pi R L D C_{A,0}}{L_d} \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)}$$

Catalyst Efficiency

$$\begin{aligned}\eta &= \frac{v}{\gamma} \\ &= \frac{\frac{2\pi RLDC_{A,0}}{L_d} \cdot \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)}}{kC_{A,0}a\pi R^2 L} \\ &= \frac{2D\beta I_1(\alpha)}{akRL_d(\beta I_0(\alpha) + I_1(\alpha))} \\ &= \frac{2}{R\sqrt{\frac{ka}{D}}} \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)} \\ &= \frac{2}{\alpha} \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)}\end{aligned}$$

Catalyst Efficiency

Assume $\beta \gg 1$

$$\beta = \frac{k_c}{\sqrt{kaD}}$$

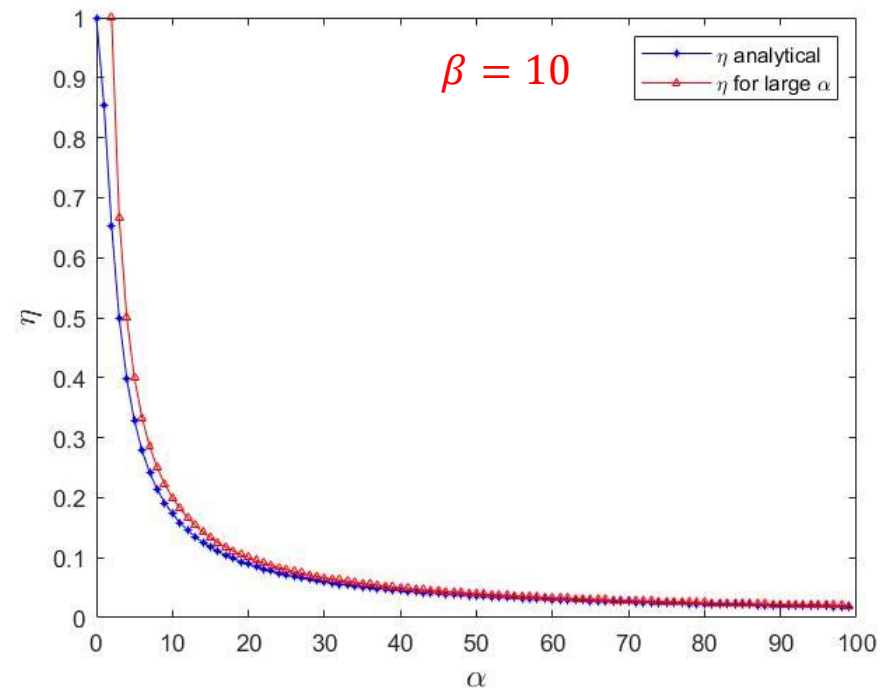
$$\eta = \frac{2}{\alpha} \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)} \sim \frac{2}{\alpha} \frac{I_1(\alpha)}{I_0(\alpha)}$$

For limiting small α :

$$\lim_{\alpha \rightarrow 0} \eta \sim \frac{2}{\alpha} \frac{I_1(0)}{I_0(0)} \sim \frac{2}{\alpha} \frac{\frac{\alpha}{2}}{1} = 1$$

For limiting large α :

$$\lim_{\alpha \rightarrow \infty} \eta \sim \frac{2}{\alpha} \frac{\frac{e^\alpha}{\sqrt{2\pi\alpha}}}{\frac{e^\alpha}{\sqrt{2\pi\alpha}}} \sim \frac{2}{\alpha}$$



Properties of Bessel Functions

Function	Small x	Large x
$J_p(x)$	$\frac{x^p}{2^p \Gamma(p+1)}$	$\sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\pi + p\pi}{4}\right)$
$Y_0(x)$	$\frac{2}{\pi} \ln x$	$\sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{\pi + p\pi}{4}\right)$
$Y_p(x), p \neq 0$	$\frac{-2^p (p-1)!}{\pi x^p}$	$\sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{\pi + p\pi}{4}\right)$
$I_p(x)$	$\frac{x^p}{2^p \Gamma(p+1)}$	$\frac{e^x}{\sqrt{2\pi x}}$
$K_0(x)$	$-\ln x$	$\sqrt{\frac{\pi}{2x}} e^{-x}$
$K_p(x), p \neq 0$	$\frac{2^{p-1} (p-1)!}{x^p}$	$\sqrt{\frac{\pi}{2x}} e^{-x}$

$$p = \frac{1}{2} :$$

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin(x)$$

$$J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos(x)$$

$$I_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sinh(x)$$

$$I_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cosh(x)$$

$$J'_0(x) = -J_1(x) \quad J_p(0) = \begin{cases} 1 & p = 0 \\ 0 & p = 1, 2, \dots \end{cases}$$

$$Y'_0(x) = -Y_1(x) \quad Y_p(0) = -\infty$$

$$I'_0(x) = I_1(x) \quad I_p(0) = \begin{cases} 1 & p = 0 \\ 0 & p = 1, 2, \dots \end{cases} \quad \& Y_p(\infty) = \infty$$

$$K'_0(x) = -K_1(x) \quad K_p(0) = \infty \quad \& \quad K_p(\infty) = 0$$

Generalized Bessel ODE

$$x^2 y'' + x(a + 2bx^r)y' + [c + dx^{2s} - b(1 - a - r)x^r + b^2 x^{2r}] y = 0; \quad \text{for } r, s \neq 0$$

with general solution in the form

$$y = x^{\frac{1-a}{2}} e^{-\frac{bx^r}{r}} \left[AZ_p \left(\frac{\sqrt{|d|}}{s} x^s \right) + BZ_{-p} \left(\frac{\sqrt{|d|}}{s} x^s \right) \right] \quad \text{where } p = \frac{1}{s} \sqrt{\left(\frac{1-a}{2} \right)^2 - c}$$

1. If $\frac{\sqrt{d}}{s}$ is *real* and p is *not* an integer (or zero), then Z_p denotes J_p and Z_{-p} denotes J_{-p}
2. If $\frac{\sqrt{d}}{s}$ is *real* and p is zero or an integer k , then Z_p denotes J_k and Z_{-p} denotes Y_k
3. If $\frac{\sqrt{d}}{s}$ is *imaginary* and p is *not* zero or an integer, then Z_p denotes I_p and Z_{-p} denotes I_{-p}
4. If $\frac{\sqrt{d}}{s}$ is *imaginary* and p is zero or an integer k , then Z_p denotes I_k and Z_{-p} denotes K_k

HW1 Q2: Solve for the dimensionless concentration

2. A water stream with a constant volumetric flow rate Q is fed into a pipe of radius R with a permeable wall, as illustrated below. The pipe is embedded in a porous medium that extends infinitely in the radial direction but has a finite height L . The water contains a dissolved reagent at an initial concentration C_F which undergoes a first-order reaction on the surface of the solid particles within the porous medium. The reaction rate is given by $r_s = kC$ ($\frac{\text{mole}}{\text{cm}^2\text{s}}$). The molar flux of the reagent N is expressed as

$$N = Cv - D \frac{dC}{dr}$$

where v is the superficial velocity and D is the effective diffusivity.

(a) Using the shell balance technique, show that the steady-state concentration distribution is governed by the equation:

$$r^2 \frac{d^2 C}{dr^2} + r \left(1 - \frac{rv}{D}\right) \frac{dC}{dr} - \frac{ak}{D} r^2 C = 0$$

where a is the specific interfacial area of the porous solid ($\frac{\text{cm}^2}{\text{cm}^3}$). Clearly state any assumptions made in the derivation. [*Hint: For dilute solutions, mass continuity requires $\frac{d}{dr}(rv) = 0$.*]

(b) Identify the appropriate boundary conditions for the governing equation. *Hint: Consider the two physical boundaries in the system.*

(c) Nondimensionalize the governing equation and its boundary conditions.

HW1 Q2: Solve for the dimensionless concentration

(c) From mass continuity condition $\frac{d(rv)}{dr} = 0$, rv is constant and hence $rv = Rv(R)$. With (12) to obtain $rv = \frac{Q}{2\pi L}$ and then substitute this into equation (10),

$$r^2 \frac{d^2 C}{dr^2} + r \left(1 - \frac{Q}{2\pi L D} \right) \frac{dC}{dr} - \frac{ak}{D} r^2 C = 0 \quad (15)$$

Let $\tilde{C} = \frac{C}{C_F}$ and $\tilde{r} = \frac{r}{R}$, the equation (15) can be transformed into its dimensionless form as

$$\tilde{r}^2 \frac{d^2 \tilde{C}}{d\tilde{r}^2} + \tilde{r} (1 - \theta) \frac{d\tilde{C}}{d\tilde{r}} - \phi \tilde{r}^2 \tilde{C} = 0 \quad (16)$$

where $\theta = \frac{Q}{2\pi L D}$ and $\phi = \frac{akR^2}{D}$. The dimensionless boundary conditions are

$$\tilde{C}(1) - \frac{2\pi L D}{Q} \frac{d\tilde{C}}{d\tilde{r}} \bigg|_{\tilde{r}=1} = 1 \quad (17)$$

$$\tilde{C}(\tilde{r} \rightarrow \infty) = 0 \text{ or } \frac{d\tilde{C}}{d\tilde{r}} \bigg|_{\tilde{r} \rightarrow \infty} = 0 \quad (18)$$

HW1 Q2: Solve for the dimensionless concentration

Announcements

Week 5 Homework

- Due Date: Week 6 class – 2:00 PM, March 4, 2026
- Required group work and intergroup discussions are encouraged

Week 6 Class

- Lab Session: Please bring your laptop with MATLAB installed
- Mid-Term Project Consultation

Week 7 Individual Meetings

- Date: Monday, March 9, 2026, Time: **1:45 PM – 4:15 PM**
- Date: Wednesday, March 11, 2026, Time: **10:00 AM – 12:30 PM**
- Action Required: Please indicate your time slot using this link

<https://docs.google.com/spreadsheets/d/10sE6iWVvFoSE9Oo5DYvNaOhwu1nqeNufKXq8NkKbHRs/edit?gid=0#gid=0>